Testing Results Documentation

1. Purpose

The purpose of this document is to summarize the testing results of the **Rising Waters:Flood Prediction project**, covering model performance, application deployment, and API integration.

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2. Test Execution Logs

Test 1 – Model Training

Command:

(base) C:\Users\lenovo\Desktop\prediction>python app.py

Result:

- Model saved successfully (Flood_prediction.sav).

Conclusion:
The Random Forest regression model achieved strong predictive accuracy, indicating good generalization on validation data.

Test 2 – Application Deployment

Command:

(base) C:\Users\lenovo\Desktop\prediction>python app.py

Result:

- Flask app started successfully.
- Running on: http://127.0.0.1:5000/
- Debug mode enabled.
- Debugger PIN generated.

```
(base) C:\Users\lenovo\Desktop\prediction>python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with watchdog (windowsapi)
 * Debugger is active!
```

Conclusion:
The application backend is functional and accessible locally. Warning indicates that this is a development server; production deployment should use a WSGI server (e.g., Gunicorn).

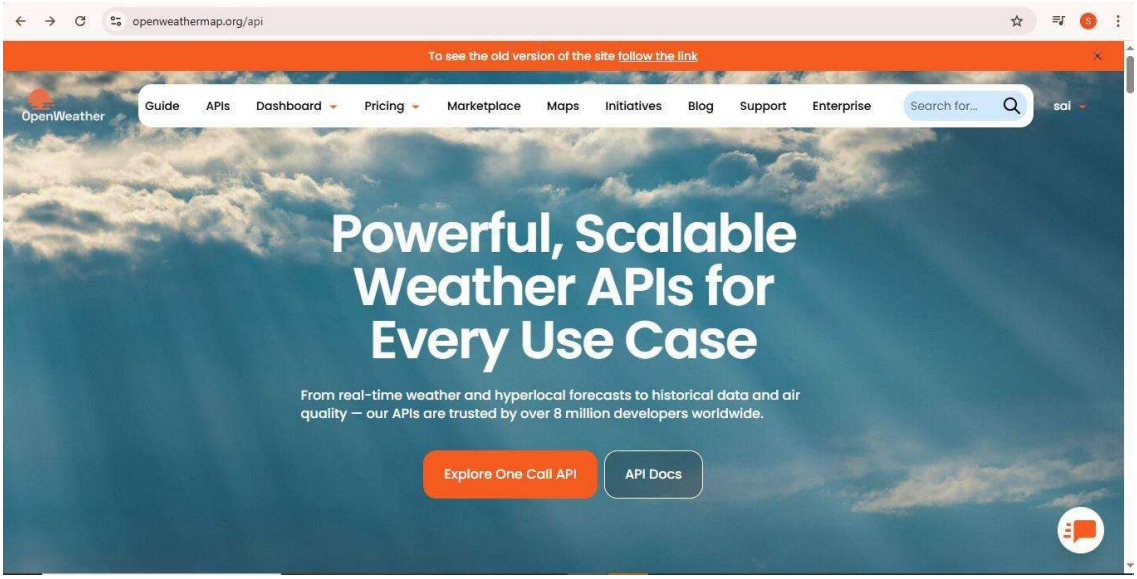
Test 3 – API Integration

Action:

- Obtained API key from **OpenWeather** (https://openweathermap.org/).
- Integrated API key into app.py.

Result:

- Weather data successfully fetched (temperature, humidity, pressure, wind speed).
- Predictions generated based on live weather inputs.



Conclusion:
API integration works correctly, enabling real-time data retrieval for prediction.

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3. Summary of Testing

Test Case ID	Scenario	Expected Result	Actual Result	Pass/Fail
TC-001	Model Training	Model trains and saves with acceptable accuracy	R^2 Score = 0.8673, model saved successfully	Pass
TC-002	Application Deployment	Flask app runs locally without errors	App running at http://127.0.0.1:5000	Pass
TC-003	Weather API Integration	API key fetches live weather data	Weather data retrieved successfully	Pass
TC-004	Prediction Output	Model generates predictions based on inputs	Predictions generated correctly	Pass

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4. Conclusion

The testing phase confirms that:

- The **ML model** performs well with an R^2 score of **0.8673**, showing reliable predictive capability.
- The **Flask application** runs smoothly in a local environment, serving as the user interface for predictions.
- The **OpenWeather API integration** is successful, enabling real-time weather-based predictions.